HealthFed-DP: Differential Privacy and Federated Learning in Healthcare

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1. Abstract & Executive Summary

Clinical artificial intelligence deployment faces critical data governance hurdles due to patient privacy laws including HIPAA and GDPR. Centralized data pooling from disparate hospital networks introduces severe legal liabilities and data leak vulnerabilities.

This paper presents HealthFed-DP, a privacy-preserving federated learning framework for training deep diagnostic classifiers on multi-site chest X-ray radiographs without sharing patient raw data.

2. Privacy Guarantees & Differential Privacy

HealthFed-DP incorporates Renyi Differential Privacy (RDP) with parameter epsilon = 1.25 and delta = 1e-5. Gradient clipping threshold C was fixed at 1.0, and Gaussian noise was injected into client weight updates before server aggregation.

Secure multi-party aggregation was executed using homomorphic encryption (Paillier cryptosystem), ensuring the central server cannot inspect individual hospital model updates.

3. Multi-Institutional Clinical Results

We deployed HealthFed-DP across 5 simulated hospital nodes comprising 42,000 chest radiography scans for pneumonia and cardiomegaly detection. The federated model achieved an Area Under the ROC Curve (AUC) of 0.932.

In comparison, a model trained on individual hospital data achieved only 0.814 AUC, demonstrating the immense statistical value of collaborative multi-institutional training without raw data migration.

4. Challenges and Regulatory Limitations

Network bandwidth constraints during homomorphic encryption aggregation introduced a 2.4x latency increase over unencrypted federated baselines.

Furthermore, non-IID data distribution across specialized regional clinics caused minor convergence instability in rare disease classes.